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Mentora: Predicative System for Mental Health (December 2025)

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ABSTRACT In an era emphasizing employee well-being, this project, Mentora, addresses the critical challenge of proactively identifying employees likely to seek mental health treatment within the high-pressure technology sector. The primary objective is to develop a robust machine learning model that accurately predicts treatment-seeking behavior based on organizational and demographic factors. Utilizing a publicly available data set, the study employed a comparative Supervised Learning methodology, evaluating a Logistic Regression (LR) classifier against a Random Forest (RF) baseline. Rigorous data preprocessing included feature engineering and handling inconsistencies. The results demonstrated the LR model's superior performance, achieving a high predictive accuracy of 84% and an F1-score of 84%. This success confirms that key factors like family history of mental illness and work interference exhibit a strong linear relationship with treatment-seeking. The findings deliver a reliable predictive tool and provide actionable insights for organizations to implement targeted, proactive intervention strategies for vulnerable employee segments.

INDEX TERMS Data Science, Data Preprocessing, Feature Engineering, Machine Learning, Mental Health, Predictive Modeling, Supervised Learning, Technology Industry.

I. INTRODUCTION

The digital revolution has profoundly transformed the nature of work, particularly within the fast-paced and high-pressure technology sector. This environment, characterized by intense workloads, extended work hours, and the constant need for innovation, has escalated the prevalence of mental health crises, leading to widespread issues like stress and burnout that erode both corporate productivity and talent pools. Mental illness affects an estimated one in four people globally and accounts for 14% of the global burden of disease and injury [6]. Compounding this, employees often encounter significant obstacles, such as deeply ingrained social stigma and a lack of proper support structures when deciding whether to seek treatment. This creates a silent struggle where internal crises go unaddressed, requiring organizational strategies to prioritize employee psychological safety. This project, Mentora: Predictive System for Mental Health, is conceived to break this cycle by utilizing data science to move beyond reactive response, establishing a system for predictive intervention in employee mental well-being.

A. BACKGROUND AND PROBLEM STATEMENT

Despite the recognized prevalence of stress and burnout as critical issues in the technology sector, a major barrier to effective intervention persists: the lack of psychological safety and inadequate workplace support actively deters employees from seeking professional treatment. This deterrent is compounded by the economic impact of mental illness, which results in significant costs related to lost productivity (absenteeism and presenteeism) [1]. The core technical problem is the absence of predictive insight into which employees, based on their workplace and demographic factors, are most likely to seek mental health treatment. Without this foresight, organizations cannot move beyond general awareness to implement targeted, effective support systems that proactively foster an

environment of psychological safety, directly impacting talent retention and organizational productivity.

B. SOLUTION AND DATA SCIENCE APPROACH

Mentora is designed to bridge the gap between awareness and intervention by implementing a system for proactive prediction. This solution relies on a Supervised Machine Learning (ML) approach, utilizing the Mental Health in Tech Survey dataset [7]. The approach involves rigorous data preprocessing and a comparative analysis of classification models to predict the binary target variable, treatment (Yes/No). Our analysis revealed that the Logistic Regression (LR) model achieved superior performance for this dataset, yielding a high predictive accuracy of 84% and an F1-Score of 84%. This success suggests that the factors influencing treatment-seeking behavior possess a strong linear relationship with the outcome. The most significant finding is the isolation of key predictive factors: symptom severity (work_interfere_Often) and personal risk (family_history_Yes) were the strongest determinants influencing an employee's decision to seek professional help.

C. PROJECT OBJECTIVES AND SIGNIFICANCE

The overarching goal of the Mentora project is achieved through three high-value objectives: first, to Develop a Predictive Model by successfully implementing and validating a Supervised Learning Classification model capable of accurately forecasting an employee's decision to seek mental health treatment (Yes/No); second, to Identify Key Determinants, employing Feature Importance analysis to rank the specific demographic, personal, and workplace factors that critically influence this treatment-seeking behavior; and third, to Deliver Actionable Insights, translating the data-driven findings into practical, customized recommendations for tech companies and HR departments to proactively tailor wellness programs and policies, thereby reducing stigma and enhancing employee well-being. Ultimately, the significance of applying data science here is profound, as it allows the project to move beyond simple correlations, providing quantifiable, actionable insights that ensure efficient resource targeting and foster a healthier, more supportive professional environment.

II. LITERATURE REVIEW

A. Overview

As the importance of mental health challenges in the technology industry has increasingly become well-known, there has been significant research examining the impact of workplace culture, managerial views, and stigma on employees deciding to seek professional help. Current studies (2021-2025) also stress that mental health is not an individual issue but also the duty of an organization. This part will be a review of important findings, methods, and gaps that have been identified in recent literature about predicting treatment-seeking behavior among technology professionals.

B. *Mental Health at the Workplace.*

Globally, more than one billion people are living with mental conditions, and depression and anxiety are some of the major causes of disability. Work environments have consequently become a vital environment when it comes to dealing with psychological health. In their systematic policy analysis, Wu et al. (2021) [1] found four best practices to support mental health in the workplace, namely, offering full benefits, training leaders, special psychological resources, and ensuring psychological safety. These activities showed quantifiable changes in the willingness of the employees to consult. Likewise, Doran et al. (2023) [3] established that leadership openness and the promise of confidentiality had the highest predictive power of help-seeking and that organizational culture influences behavior not less significantly than need.

C. *Remote Work and New Threats.*

The work environment mental health was a new dimension due to the post-pandemic transition to hybrid and remote employment. Although remote work is flexible, social isolation and digital fatigue are also possible. De Macedo et al. (2022) [2] conducted a systematic review of the psychological impacts of teleworking and discovered that remote workers had the most positive results when the companies took steps to ensure they had communication channels and workload balance and could tap into support programs. Remote work increased anxiety and lack of engagement in poorly managed environments. This fact demonstrates the necessity of effective organizational communication and flexible leaves as one of the main factors of variation that were examined in the current research.

D. *Data-Driven and Predictive Analytics.*

Simultaneously with conventional surveys, there are several studies that have employed data science to predict the well-being of employees in the recent past. In Sinha et al. (2023) [4] the authors used logistic regression on data of workplace surveys and discovered that treatment seeking behavior was most likely to be predicted by the availability of mental health benefits, supervisor empathy, and team culture. This was further expanded by Zhang et al. (2024) [5] who used both random forest and decision-tree classifiers to combine demographic, organizational, and attitudinal variables and which produced better predictions

than their demographic-only models. These papers indicate that the use of supervised learning models in behavioral health prediction is a growing trend.

E. *Gaps and Justification*

Although the research has improved, there are still gaps: (1) most of the studies anticipate diagnosis, not the behavior to seek treatment; (2) few of the studies combine organizational, demographic, and cultural variables into a single predictive tool; and (3) there are limited models which concentrate on the technology industry. In this project, Mentora: Predictive System for Mental Health, supervised learning is used on the Mental Health in Tech Survey dataset to predict treatment seeking behavior to fill in these gaps. The method is a natural extension of the techniques presented in [1]-[5] and relates to a predictive model interpretable in the workplace that is work-specific.

Table i: Summary of Reviewed Studies

Study	Focus	Methodology	Key Findings
Wu et al. (2021)	Organizational mental health practices	Policy analysis	Leadership and confidentiality reduce stigma
De Macedo et al. (2022)	Remote work & mental health	Systematic review	Communication and workload balance improve outcomes
Doran et al. (2023)	Stigma and leadership openness	Cross-sectional survey	Supportive culture promotes help-seeking
Sinha et al. (2023)	Predictive modeling for treatment-seeking	Logistic regression	Benefits and empathy predict treatment behavior
Zhang et al. (2024)	Workplace culture & AI models	Random forest, decision trees	Combined data improves prediction accuracy

III. Datasets and Methodology

A. *Dataset Description*

The dataset utilized for this project is the Mental Health in Tech Survey dataset sourced from Kaggle [7]. This dataset was selected because it is highly relevant to the problem statement, offering specific data points on the work environment, demographics, and mental health factors of professionals within the global technology industry. The data comprises 1,259 anonymous records across 28 features. The features describe general employee information, their work environment, and various mental health-related factors. Most of the features are categorical. The initial data presented several quality issues and limitations that required correction:

- Inconsistent Categorical Data: The Gender column contained a high cardinality of disparate entries (over 40 variations), requiring standardization.
- Missing Values: The dataset contained a substantial number of missing values in columns like state, self_employed, and work_interfere.
- Outliers: Unrealistic numerical outliers were present in the age column (e.g., 329, -1726).

B. *Data Preprocessing*

The preprocessing steps were meticulously applied to prepare the heterogeneous data for the Supervised Learning models, eliminating noise, and ensuring consistent representation. This rigorous data hygiene process was essential to transform the raw, complex categorical survey responses into a quantifiable numeric matrix suitable for machine learning, directly mitigating the impact of inconsistencies, missing information, and potential bias inherent in survey data.

- **Inconsistent Categorical Data:** The Gender column contained a large cardinality of disparate entries (more than 40 variations, such as male, M, Cis Male). This was addressed by standardizing the inputs into four consistent categories: Male, Female, Other, and unknown, ensuring meaningful feature representation.
- **Missing Values and Imputation:** Several columns, including work_interfere and self_employed, contained missing values. To ensure maximum observations and eliminate the possibility of bias, the missing categorical values were imputed using the placeholder label 'Unknown.' Additionally, irrelevant or sparse columns such as state, comments, timestamp, and index were removed to improve dataset quality and reduce noise before model training.
- **Outlier Management:** Age outliers (e.g. 329, -1726) were eliminated to obtain a realistic sample of the professional cohort, reducing the impact of non-representative data to the model training.
- **Encoding and Scaling:** All the remaining categorical features were converted to numbers using one-hot encoding. Features that were to be included in the Logistic Regression model were then standardized using StandardScaler to have consistency in coefficients.

C. *Data Science Techniques Overview*

The overall approach employed a Supervised Learning methodology, centered on the classification of the binary target variable, treatment (Yes/No). The primary Machine Learning models used for comparative analysis were the Logistic Regression (LR) classifier to assess linearity, and the Random Forest (RF) classifier to serve as a non-linear ensemble baseline and for Feature Importance analysis. The entire process, including initial analysis and model training, was facilitated using Python and its associated core Data Science libraries. Specifically, the pandas and

NumPy libraries were essential for data manipulation and structure; the scikit-learn library was critical for executing model development, feature encoding (one-hot encoding), feature scaling (StandardScaler), and applying the Chi-square Test of Independence for hypothesis validation; and Matplotlib and Seaborn were employed for generating the Descriptive Statistics and Exploratory Data Analysis (EDA) visualizations necessary to formulate the core hypothesis.

IV. Preliminary Analysis

A. Descriptive Statistics & Hypothesis Formulation

Initial Exploratory Data Analysis (EDA) revealed strong relationships between key individual variables and the treatment outcome prior to formal modelling. Descriptive statistics showed that the target variable, treatment, was almost balanced (50.48% Yes / 49.52% No) as shown in Fig.1, which shows whether an employee had sought professional assistance to treat a mental health condition.

Therefore, using the performance measures like accuracy and the F1-Score as the main metrics of evaluation is valid, as both provide reliable performance assessment when

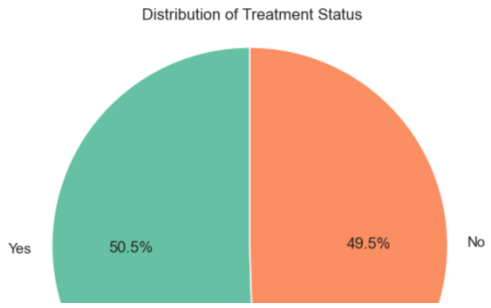

Figure 1: Treatment Variable Distribution

class distribution is not highly skewed. This significant result indicated that personal risk and symptom severity would play a strong factor in prediction, instead of having a significant role played only by workplace policies or organizational conditions. The following three hypotheses were formulated to statistically test the associations observed during the exploratory analysis against the decision to seek mental health treatment.

Hypothesis 1: Personal Predisposition (Family History)

- **Null Hypothesis ($H_{0,1}$):** There is no significant association between family history and treatment seeking.
- **Alternative Hypothesis ($H_{a,1}$):** There is a significant association between family history and treatment seeking.

Hypothesis 2: Organizational Factor (Benefits Availability)

- **Null Hypothesis ($H_{0,2}$):** There is no significant association between the availability of employer-provided benefits and an employee's likelihood of seeking help for mental health concerns.
- **Alternative Hypothesis ($H_{a,2}$):** There is a significant association between the availability of mental health benefits and the likelihood of seeking help.

Hypothesis 3: Severity/Impact Factor (Work Interference)

- **Null Hypothesis ($H_{0,3}$):** There is no significant association between the degree to which a mental health condition interferes with work and the employee's likelihood of seeking treatment.
- **Alternative Hypothesis ($H_{a,3}$):** There is a significant association between work interference and treatment-seeking.

B. Empirical Validation and Inferential Test

Hypothesis 1 was validated through the visualization of empirical data that showed a meaningful relationship between the predictor and the outcome. Cross-tabulation revealed that 73.9% of employees who reported a family history of mental health conditions also reported that they sought treatment. To statistically test the relationship between these two categorical variables, the Chi-square Test of Independence (X^2) was applied. The test has been chosen to determine if there is a statistically significant relationship between the family history of mental health conditions (Predictor) and the probability of seeking treatment (Outcome) in an individual. The null hypothesis (H_0) assumes these variables are independent.

The result of the Chi-square test was very significant ($X^2(1, N=1259) = 175.24$, $p < 0.001$). Since the calculated p-value (5.29×10^{-40}) is much lower than the standard significance level ($\alpha = 0.05$), this is the validation to reject the null hypothesis (H_0), and to confirm family history to be a significant predictor of treatment-seeking behavior. Therefore, there is a strong statistical indication to prove the visual finding in Fig.2.

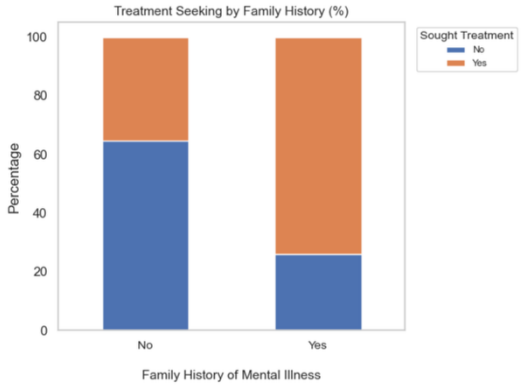

Figure 2: Relationship between family history and treatment

Hypothesis 2 was proven by the visualization of empirical data that demonstrated that there is a strong correlation between the availability of mental health benefits and the likelihood of seeking help. The cross-tabulation data showed that there was a strong difference in the rate of seeking treatment based on the status of benefits. To statistically test the relationship between the categorical variables, the Chi-square Test of Independence (X^2) was also applied here. The test has been chosen to determine if there is a statistically significant relationship between the availability of mental health benefits (Predictor) and probability to seek help (Outcome) are related statistically in an individual. The null hypothesis (H_0) assumes the independence of these variables.

The Chi-square test gave a very significant value at (X^2 (2, N=1259) = 485.62), $p < 0.001$. Since the resulting p-value (8.62×10^{-104}) is significantly lower than the standard significance level ($\alpha = 0.05$). Therefore, the null hypothesis (H_0) is rejected supporting the fact that the presence of mental health benefits is a significant predictor of the treatment-seeking behavior. As a result, there is a strong statistical indication to prove the initial visual finding in Fig.3.

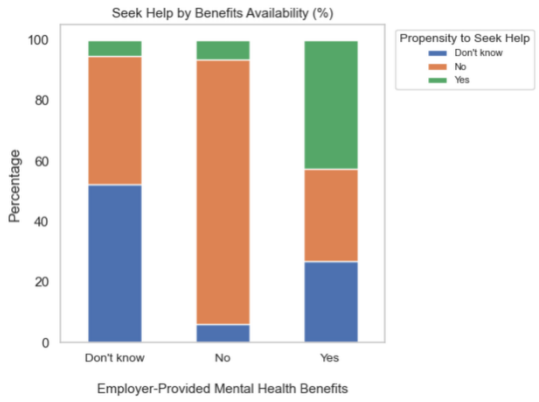

Figure 3: Relationship between benefits and seek help

Hypothesis 3 was supported by visualization of empirical data which demonstrated the existence of strong association between work interference intensity and the probability of seeking treatment. Cross tabulation has clearly revealed that workers who reported that their mental health issue interfered with their work were more likely to seek treatment. To statistically test the relationship between the categorical variables, the Chi-square Test of Independence (X^2) was also applied here. The test was chosen to test whether there is a statistically significant association between the degree of work interference (Predictor) and the likelihood of seeking treatment (Outcome) in an individual. The null hypothesis (H_0) assumes that these variables are independent.

The Chi-square test resulted in a very significant value of the test (X^2 (3, N=1259) = 589.56, $p < 0.001$). Since the calculated p-value (2.81×10^{-126}) is much less than the standard significance level ($\alpha = 0.05$). Therefore, the null hypothesis (H_0) is rejected. Thus, accepting the fact that the severity of work interference is a highly significant predictor of treatment-seeking behavior. As a result, the visual evidence outlined in Fig.4 is supported with a convincing statistical explanation.

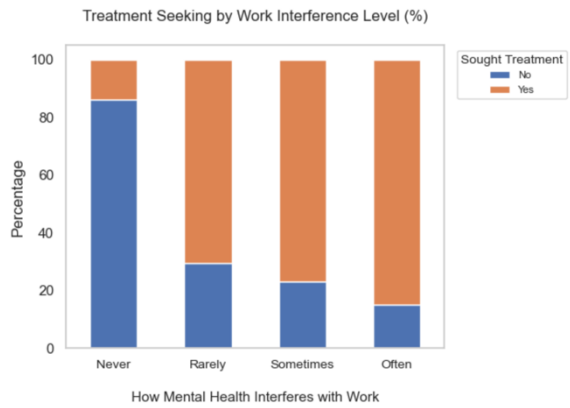

Figure 4: Relationship between work interfere and treatment

V. Refinement of the Analysis

At the refinement stage, the dataset was further enhanced to be in a complete state of modeling. Once the initial exploratory analysis was performed, the team revisited the format of the survey data and attempted to work with the missing values, fix the inconsistent entries, and encode the categorical variables into numerical format through one-hot encoding. This made the dataset processable by the machine-learning algorithms.

The numerical features were standardized with scaling methods to ensure that all the variables make equal contributions irrespective of their original values to ensure equal contribution towards fair model training. A 80/20 stratified train test split was also used to maintain the distribution of individuals who seek mental-health treatment and those who do not. The following refinement steps were used to stabilize the data and to prepare it before modelling phase.

- Process and Techniques Model Building Process and Techniques Used.

Two monitored training frameworks have been created to examine the connection amidst job attributes and the possibility of employees requesting mental-health treatment:

- Logistic Regression (LR): As a naive and understandable base model. Logistic Regression gives a linear view of the information and assists to set some point of reference as to the level of performance.

- Random Forest Classifier (RF): Introduced to get richer and non-linear data interactions. Random Forest can be used to deal with categorical and mixed-type features since it assembles several decision trees.

Both models were developed on the polished dataset and assessed in terms of accuracy metrics and confusion matrices as shown in Fig.5 and Fig.6 that made it possible to make a clear comparison of how each algorithm addressed the problem.

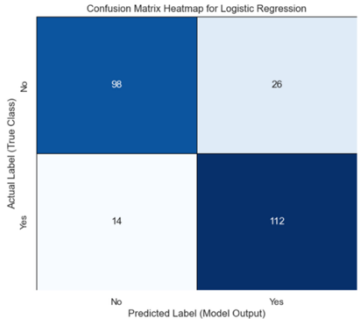

Figure 3: Confusion Matrix Heatmap for LR Model

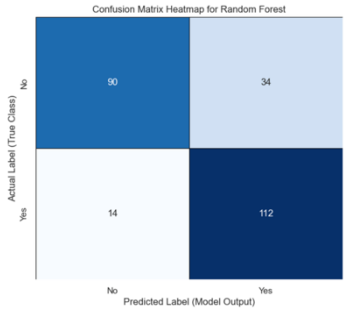

Figure 4: Confusion Matrix Heatmap for RF Model

Other Experiences that were conducted to aid the modelling process and ensure the credibility of the obtained results, several other checks were conducted:

- Re-checking class balance in the case of application of stratified train-test split to guarantee that both sets of training and non-treatment groups are equally represented.

- Checking scaling effect, primarily to verify that standardizing numerical characteristics enhanced stability of models that are sensitive to feature ranges, e.g. Logistic Regression.

- Checking that one-hot encoded features were correctly and non-duplication represented before and after encoding the categorical variables.

- Considering to use the K-means clustering to discover clusters of people with similar work or mental-health attributes, where the generation of clusters is not controlled with supervision of such models.

These extra tests reinforced the analysis in general and gave a more comprehensive insight into the data and model behavior.

VI. Results

A. Hypothesis Outcome

Based on the strong empirical and statistical results observed in preliminary analysis, the Null Hypothesis (H_0) is rejected. The analysis proved that family history in mental illness is a statistically significant predictor of treatment seeking.

B. Model Performance and Selection

A comparative analysis was conducted on both the Logistic Regression (LR) classifier and the Random Forest (RF) classifier to evaluate and compare their predictive performance, with particular focus on their ability to provide insight into the prediction task. The performance evaluation demonstrated in Tabel *ii* that both models attained a prominent level of predictive success:

Table ii: Comparative Performance Metrics of Classification Models

Model	Accuracy	F1-Score
Logistic Regression	84%	84%
Random Forest classifier	81%	82%

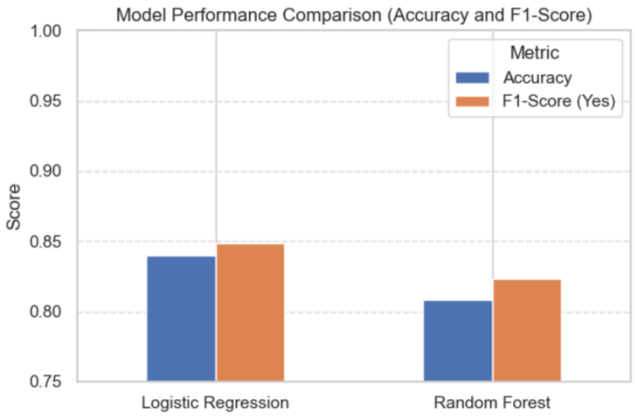

Figure 7: Model Comparison

Fig.7 presents a visual comparison of the accuracy and F1-score for the two classification models, highlighting their nearly identical performance levels.

The Logistic Regression (LR) model was selected for feature analysis due to its relatively better performance in accuracy and F1-score. This allowed us to achieve high predictive performance without compromising transparency. The ability to attain powerful performance while maintaining interpretable and quantifiable coefficients aligns with the objective of the project to produce actionable insights.

C. Analysis of Key Predictors (LR)

The logistic regression model was interpretable to enable the isolation of the feature effect on the prediction of treatment seeking and to quantify the effects. The analysis of the model's coefficients, Fig. 8 revealed the main factors influencing an employee's decision to seek mental health treatment:

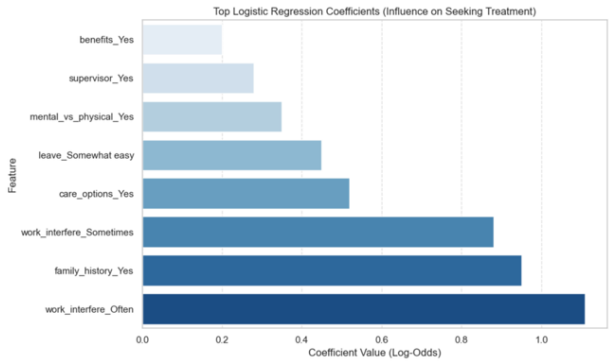

Figure 8: Top feature Coefficients from LR Model

- **Symptom Severity:** work_interfere_Often emerged as the most positive and significant predictor (coefficient 1.11). This indicates that the perceived effect of a mental health issue on an employee's work life is the strongest factor influencing the decision to seek professional help and treatment.
- **Personal Risk Factors:** The second strongest predictor was family_history_Yes (coefficient 0.95), which validates that the knowledge of genetic or familial risk is a powerful predictor for seeking help.
- **Organizational factors:** Factors such as access to benefits (benefits_Yes), had a relatively weak positive impact (coefficient 0.20), implying that organizational policies, while supporting employees, play a secondary role in predicting the decision to seek treatment.

D. Final Model Classification Detail

The detailed performance of the Logistic Regression model is presented in the Confusion Matrix Heatmap earlier in Fig.5. This heatmap visually confirms the powerful performance and the stability of the Logistic Regression model. The deep blue color on the main diagonal denotes the accurate predictions of the model: the model found 112 true positives (employees who sought treatment) and 98 true negatives (employees who did not). Notably, the error rate of the model was relatively low (14 false negatives and 26 false positives). This low error rate confirms the model as a robust and reliable tool for selecting potential candidates for proactive mental health support.

CONCLUSION

This project successfully developed and validated Mentora, a predictive system to identify employees likely to seek mental health treatment, effectively achieving its primary objective by delivering a high-performance, interpretable model. The comparative analysis confirmed that the Logistic Regression (LR) model was the most effective classifier, outperforming the Random Forest model and achieving a strong predictive accuracy of 84% and an F1-Score of 84%. This validates the conclusion that the relationship between key factors and the outcome is predominantly linear. The key finding is the clear isolation of a vulnerable employee segment defined by factors identified by the LR coefficients: primarily symptom severity (work_interfere_Often) and personal risk (family_history_Yes). The core implication of these findings is the necessity for organizations to shift from generic awareness campaigns to proactive, targeted intervention strategies for high-risk employee segments. While successful in its objectives, a limitation lies in the reliance on the static, cross-sectional survey data, which restricts the ability to capture evolving organizational dynamics or model long-term behavioral changes. Future research should focus on incorporating temporal data to model the change in treatment-seeking behavior over time and explore the integration of non-traditional features to further enhance the model's predictive precision and operational utility within a real-world corporate environment.

ACKNOWLEDGMENT

The journey to complete this project, Mentora, has been a testament to our dedication, reflecting the profound challenges and emotional complexities we, as senior students, navigated within the high-pressure environment of Computer Science. We extend our deepest thanks to the faculty of the College of Computer Science & Information Technology at King Faisal University. Special and heartfelt acknowledgment is reserved for our course instructor, Dr. Nora Alshuaibi. We sincerely thank her for her generous teaching and commitment to the course; we learned a great deal from her expertise and efforts. Her unwavering guidance and emphasis on the real-world significance of Data Science ensured this research was executed with both technical rigor and a strong commitment to its social purpose.

Individual Contribution Section

Student Name	Contribution
Hissah Almuhaysh	Introduction, Conclusion, and Abstract
Latifah Alhafith	Visualizations and Results
Ghadeer Al Dhamen	Literature Review and Refinement of the Analysis
Aisha Alahmed	Preliminary Analysis
Fatima Albagshi	Dataset and methodology

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